

miniLISA: Design and Component Noise Model of an Optical Time-Delay Interferometry Testbed

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We describe miniLISA, a laboratory testbed combining optical laser signals, FPGA-based electronic propagation delays, and independently referenced spacecraft phasemeters. The design reproduces the laser-noise structure of LISA’s one-way measurements, a reduced sideband clock-transfer model, and electronically injected gravitational-wave phase responses. We relate the hardware signals to the interferometric measurement model and propagate measured electronic and sideband-difference spectra together with a prospective detector-noise calculation through a static, clock-corrected Michelson observable. The resulting component budget is conditional on ideal laser- and clock-noise cancellation and on the assumed origin of the sideband-difference noise. We identify delay knowledge, differential board timing, and clock-transfer characterization as remaining requirements for testing the assembled optical system, and outline the extension to time-dependent delays.

I. INTRODUCTION

Gravitational waves provide a unique observational window into astrophysical and cosmological phenomena. The first direct detection in 2015 established gravitational-wave observations of compact binary mergers with ground-based interferometers [1]. Accessing gravitational waves at millihertz frequencies requires a space-based detector to overcome terrestrial seismic and Newtonian noise and to provide sufficiently long interferometric baselines. The Laser Interferometer Space Antenna (LISA), a space-based observatory led by ESA with NASA participation, is designed to operate in the 0.1 mHz to 1 Hz band [2].

The LISA constellation consists of three spacecraft arranged in an approximately equilateral triangle with arm lengths of about 2.5 million km. Laser beams exchanged between the spacecraft form a constellation-scale interferometer. Unlike ground-based detectors with effectively equal arms, the LISA arm lengths vary by up to about 1.2% due to orbital motion, preventing direct cancellation of laser frequency noise. Although cavity pre-stabilisation reduces laser frequency noise substantially, it remains several orders of magnitude above the displacement noise budget required for gravitational-wave detection [3]. To suppress this noise, time delay interferometry (TDI) has been developed as a post-processing technique that synthesizes virtual equal-arm interferometers by appropriately time-shifting and combining phasemeter measurements.

In addition to laser frequency noise, LISA measurements are affected by clock noise originating from the independent ultra-stable oscillators (USOs) carried by each spacecraft. Since all phase measurements are referenced to these clocks, timing jitter couples directly into the measured heterodyne beatnotes. Space-qualified USOs achieve Allan deviations of order 10^{-13} at one-second averaging times [4], which is insufficient for the picometer-level test-mass position sensitivity required by LISA. To mitigate this effect, LISA employs additional clock-

noise transfer measurements—implemented via modulation sidebands exchanged between spacecraft.

TDI has been developed analytically and tested through numerical simulations [3, 5]. Simulations predict the response of a specified measurement model and test its numerical implementation. Physical experiments provide a complementary test of whether the hardware follows that model, including couplings and processing effects that an idealized simulation may omit.

The University of Florida Laser Interferometry Simulator (UFLIS) demonstrated TDI and TDI-ranging with time-varying electronic propagation delays and optical laser signals [6]. miniLISA continues this approach with FPGA delay lines and independent spacecraft clock references. The experiment of de Vine et al. demonstrated laser-noise suppression of approximately 10^9 and clock-noise suppression of 6×10^4 , using short optical paths [7]. LISA On Table demonstrated first-generation TDI with simulated noise and static unequal arms [8].

Here, we present an overview of an ongoing effort to develop a TDI testbed that combines realistic laser frequency noise, programmable delay lines, and independent clock sources within a single experimental framework. The testbed focuses on reproducing the dominant noise couplings of the LISA metrology system—specifically, laser frequency noise and clock noise—with a static-delay performance model and a planned extension to unequal, time-dependent delays. Crucially, the same delay lines that impose the interspacecraft light-travel times also allow physically consistent gravitational-wave strain signals to be injected directly into the measurement chain. The testbed can therefore be used not only to demonstrate noise suppression, but to verify that a known gravitational-wave signal survives TDI processing of hardware-generated data with the expected amplitude and phase.

An initial purely electrical implementation centered around field programmable gate array (FPGA)-based digital delay lines has already been demonstrated in Ferguson et al. [9]. The developments discussed here extend that architecture by incorporating an optical front-

end with physically representative laser noise and independently running spacecraft clocks. This reproduces the laser-noise and propagation structure of the one-way measurements together with a reduced clock-transfer model, allowing TDI combinations and clock-noise calibration to be formed entirely from hardware-generated data. The goal of this work is to realize a laboratory-scale analog of the LISA inter-satellite metrology system—a “miniLISA”—that bridges the gap between idealized software simulations and the flight instrument, and provides a controlled environment for testing the robustness of TDI to instrumental non-idealities and realistic gravitational-wave signal injection.

More concretely, we look for residual laser and clock noise below the secondary-noise floor after processing, together with recovery of an injected GW signal at the expected amplitude and phase.

Section II introduces the LISA measurements and noise-suppression combinations. Section III describes their laboratory implementation, and Section IV estimates its performance from measured and calculated component noises. We then discuss possible extensions and remaining measurements.

II. LISA INTERFEROMETRIC DETECTION SYSTEM

We now describe the LISA interferometric detection system (IDS) and present a mathematical model of the noises. We specifically focus on relevant elements for laser frequency noise and USO jitter propagation and coupling.

A. Interferometric Phase Measurements

Each of the three identical LISA satellites carries two lasers on board. We refer to one as the local laser and the other as the adjacent laser. These two laser fields interfere on the test-mass and reference interferometers on the two optical benches of the spacecraft. In addition, the local laser interferes with the laser received from a distant spacecraft on the science interferometer. It is this science interferometer that encodes the gravitational-wave signal by measuring fluctuations in the spacecraft-to-spacecraft distance, while the test mass interferometer will measure the spacecraft-to-test mass displacement. In the upper panel of Fig. 1, a simplified LISA two-way armlink involving only the science interferometer is shown.

We begin by writing the optical phase of the laser field originating from movable optical system assembly (MOSA) ij on spacecraft i facing spacecraft j as a function of time t :

$$\Phi_{ij}(t) = \nu_{ij}t + \delta\phi_{ij}(t), \quad (1)$$

where ν_{ij} denotes the nominal carrier frequency and $\delta\phi_{ij}(t)$ represents the intrinsic laser phase noise.

Throughout this section, phases are expressed in cycles, frequencies in Hz, and clock errors in seconds. Capital Φ denotes total phase, while $\delta\phi$ denotes phase fluctuations about the nominal linear evolution. We use a single time coordinate, neglect relativistic time transformations, and retain constant nominal frequencies in the noise-coupling coefficients. The clock on spacecraft i reads $t_i = t + q_i(t)$, where q_i is its timing error.

Each of the laser fields is phase-modulated using an electro-optic modulator (EOM) driven by the local USO. This is used to imprint sidebands on the laser field that carry the USO timing noise $q_i(t)$, creating a phase noise term $\nu_{ij}^m q_i(t)$. Since there is only one USO per spacecraft, clock noise carries a single spacecraft index.

The modulation frequency ν_{ij}^m will be about 2.4 GHz, which is upconverted from the USO via a phase locked loop (PLL) synthesizer, preserving its noise with high fidelity. For the adjacent and opposing MOSAs ik and ji respectively, the corresponding modulation frequency differs by 1 MHz, so 2.401 GHz, such that the sideband-sideband beat notes are offset by 1 MHz from the carrier-carrier beat note. The modulation depth $\mu_{\text{EOM}} = 0.53$ rad places about 87% of the optical power in the carrier and 6.5% in each first-order sideband. The USO references the modulation, sampling-clock and pilot-tone synthesis chains; their role in clock-noise suppression is discussed in Section IIC. Additional phase modulation with pseudo-random codes supports ranging and data transfer; it is distinct from the sinusoidal clock-transfer modulation considered here [10].

Propagation from MOSA ij to ji delays the emitted phase and adds the GW response along the arm. Following the separation of background propagation and GW perturbations in [3], we write

$$d_{ji}^{\text{tot}}(t) = d_{ji}(t) + H_{ji}(t). \quad (2)$$

Here $d_{ji}(t)$ is the background propagation time, which can vary slowly with the constellation breathing, and $H_{ji}(t)$ is its GW-induced perturbation. The first index denotes the receiving spacecraft and the second the emitting spacecraft. We define the delay operator using only the background propagation time,

$$\mathbf{D}_{ji}x(t) = x(t - d_{ji}(t)). \quad (3)$$

Applying the total propagation time to Eq. (1) and expanding to first order gives

$$\begin{aligned} \Phi_{ij}(t - d_{ji}(t) - H_{ji}(t)) &\simeq \nu_{ij} [t - d_{ji}(t)] \\ &\quad + \mathbf{D}_{ji}\delta\phi_{ij}(t) - \nu_{ij}H_{ji}(t), \end{aligned}$$

where the product $H_{ji}\mathbf{D}_{ji}\dot{\delta\phi}_{ij}$ of the GW perturbation and laser-frequency noise has been neglected. We therefore introduce the GW phase contribution

$$\delta\phi_{ji}^{\text{GW}}(t) = -\nu_{ij}H_{ji}(t). \quad (4)$$

To this order, the received total phase is $\mathbf{D}_{ji}\Phi_{ij}(t) + \delta\phi_{ji}^{\text{GW}}(t)$: the GW signal is an additive phase contribution, separate from the background delay operator. For

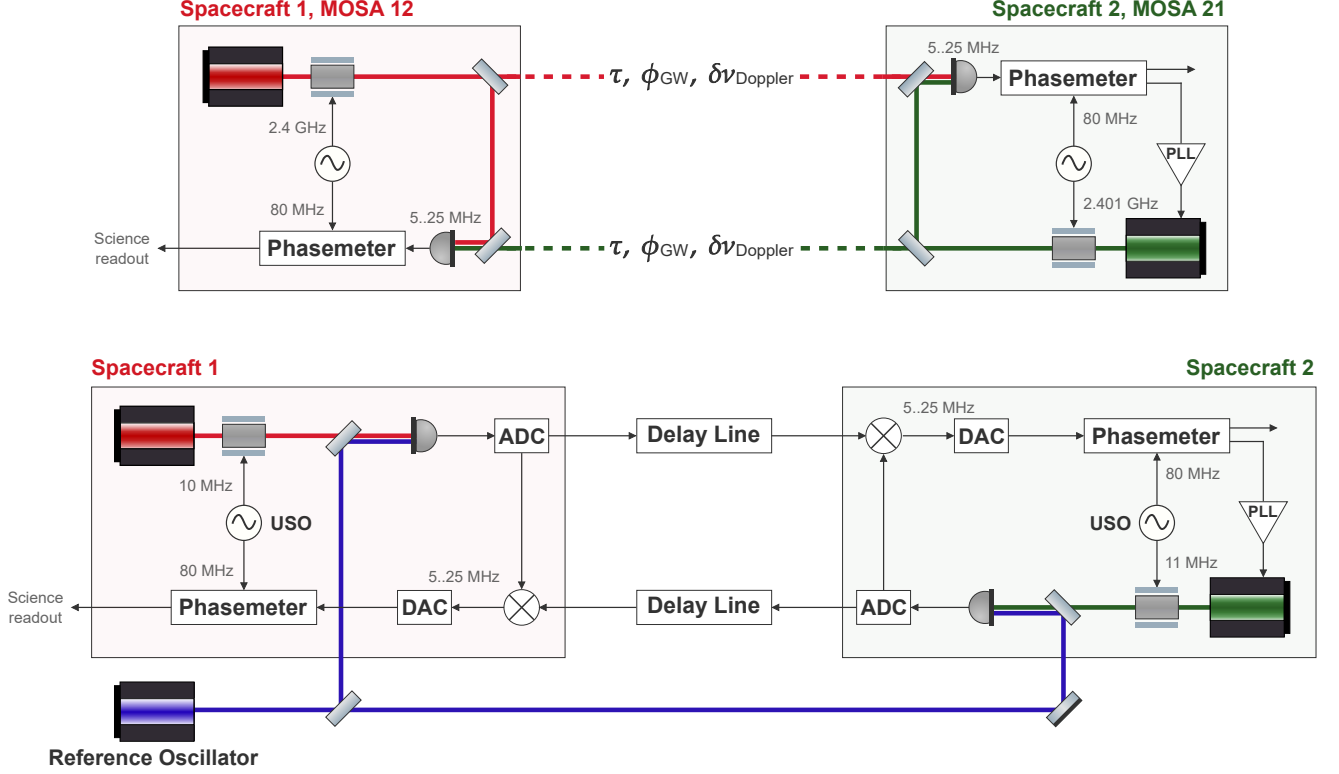


FIG. 1. A comparison of the two-way links as implemented in LISA (upper) and the testbed (lower). The upper panel shows the conceptual LISA configuration: the laser on Spacecraft 1 (MOSA 12) is phase-modulated at approximately 2.4 GHz using the local USO and transmitted to Spacecraft 2 (MOSA 21). The received beam interferes with the local laser, producing heterodyne beat notes in the MHz range that contain gravitational-wave phase fluctuations, Doppler shifts due to orbital motion, and laser frequency noise. These beat notes are tracked by the phasemeter, which is clocked by the onboard 80 MHz sampling reference derived from the USO. The lower panel depicts the laboratory implementation of the same signal topology. Instead of physical light propagation over millions of kilometres, the inter-spacecraft delay is emulated electronically using digital delay lines between ADC and DAC stages. A common reference laser is used to downconvert the laser fields from hundreds of terahertz to tens of megahertz, which can then be sampled by the delay line ADCs. The actual LISA-like heterodyne beat notes are created in the mixers.

Draft note: Revise the lower-panel labels to independent Rb-referenced Mokus, (35, 36, 34) MHz modulation and the synthesized 3–8 MHz output band. The legacy USO and 10/11 MHz labels do not describe the operating point used here.

constant ν_{ij} , the corresponding frequency fluctuation is $\delta\dot{\phi}_{ji}^{GW} = -\nu_{ij}\dot{H}_{ji}$.

At spacecraft j , the received field i interferes with the local laser field of phase $\Phi_{ji}(t)$ on a photodetector. The resulting heterodyne signal contains both the carrier–carrier beat note and the sideband–sideband beat notes. Additional cross terms lie outside the heterodyne range of the phasemeter; a representative LISA frequency plan uses absolute beat frequencies from 5 to 25 MHz [11].

The electrical signal from the photodetector is then digitized by the phase measurement system (PMS) front-end electronics. The sampling process introduces a coupling of the local USO timing error $q_j(t)$ into the measured phase. For all measurement readouts, we adopt the LISA convention that phases are written as “distant

minus local.”¹ We define the nominal carrier coefficient

$$\alpha_{ji} = \Delta\nu_{ji} = \nu_{ij} - \nu_{ji}. \quad (5)$$

After removing the nominal beat evolution, the sampled carrier phase fluctuations are

$$\delta\phi_{ji}^c(t) = \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) + \delta\phi_{ji}^{GW}(t) - \alpha_{ji}q_j(t). \quad (6)$$

Here α_{ji} is the beat frequency in the fixed-frequency approximation used for the noise model. For an orbit-dependent frequency plan, it would be replaced by the actual signed beat frequency, including Doppler shifts.

¹ The phasemeter measures the absolute beat frequency; its sign is assigned using the frequency plan.

The signal is then passed to digital phase lock loops (DPLLs), which track the carrier and both sideband beat notes. In the following readouts, the nominal phase evolution is largely removed from the tracked phase, leaving the fluctuations used for GW detection and clock-noise suppression. To connect this noise model to the measurements used for TDI, we also include the secondary disturbances in the science readout:

$$\begin{aligned} \text{sci}_{ji}^c(t) = & \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \alpha_{ji} q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^c(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji} \mathbf{\Delta}_{ij}(t) + \mathbf{n}_{ji} \cdot \mathbf{\Delta}_{ji}(t)), \end{aligned} \quad (7)$$

The term $N_{ji}^c(t)$ represents combined readout noise from optical path-length fluctuations and electronics. The vectors $\mathbf{\Delta}_{ij}(t)$ are optical-bench displacements in metres. The unit vector $\mathbf{n}_{ji}$ points along the incoming beam from spacecraft i to spacecraft j , and $\mathbf{n}_{ij} = -\mathbf{n}_{ji}$ in the geometry used for these displacement terms. The factor $\nu_0/c = 1/\lambda_0$ converts path-length fluctuations to phase in cycles, with $\nu_0 \simeq 281$ THz for the nominal wavelength $\lambda_0 = 1064$ nm. MHz-scale offsets between lasers are small compared with this optical frequency, so a common ν_0 suffices for the displacement terms. Those offsets remain essential in the heterodyne clock coefficients, which depend on frequency differences.

The corresponding upper sideband–sideband beat note readout can be written as

$$\begin{aligned} \text{sci}_{ji}^{\text{sb},+}(t) = & \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \gamma_{ji} q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^{\text{sb},+}(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji} \mathbf{\Delta}_{ij}(t) + \mathbf{n}_{ji} \cdot \mathbf{\Delta}_{ji}(t)) \\ & + \nu_{ij}^m \mathbf{D}_{ji} q_i(t) - \nu_{ji}^m q_j(t). \end{aligned} \quad (8)$$

The sideband clock coupling coefficient is

$$\gamma_{ji} = \Delta\nu_{ji} + \nu_{ij}^m - \nu_{ji}^m = \Delta\nu_{ji} + \Delta\nu_{ji}^m = \Delta\nu_{ji}^{\text{sb},+}. \quad (9)$$

Again, the term $N_{ji}^{\text{sb},+}(t)$ represents stochastic readout noise specific to the upper sideband measurement. The lower sideband has the same laser phase noise but the opposite sign for each modulation contribution. The sidebands have optical frequencies $\nu_{ij} \pm \nu_{ij}^m$, so the same path-length change produces a slightly different phase shift in each sideband. Using ν_0/c for their displacement couplings neglects a fractional correction of order $\nu^m/\nu_0 \simeq 9 \times 10^{-6}$. Similarly, the upper-sideband GW phase is $-(\nu_{ij} + \nu_{ij}^m)H_{ji}$, differing from the carrier by $-\nu_{ij}^m H_{ji}$. We neglect this small difference in the readout equations above and use the same $\delta\phi_{ji}^{\text{GW}}$ for the carrier and both sidebands. The δ denotes a perturbation to the nominal optical phase, rather than stochastic noise.

For completeness, we also write out the reference and

test mass interferometer readings on MOSA ji

$$\begin{aligned} \text{tm}_{ji}(t) = & \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji} q_j(t) + \mu_j(t) + N_{ji}^{\text{tm}}(t) \\ & + \frac{2\nu_0}{c} \mathbf{n}_{ji} \cdot (\mathbf{\delta}_{ji}(t) - \mathbf{\Delta}_{ji}(t)), \end{aligned} \quad (10)$$

$$\begin{aligned} \text{ref}_{ji}(t) = & \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji} q_j(t) \\ & + \mu_j(t) + N_{ji}^{\text{ref}}(t). \end{aligned} \quad (11)$$

Here we have introduced the fiber noise μ_j coming from the optical fiber that exchanges the local and adjacent laser fields between MOSA ji and MOSA jk . This term is typically referred to as the backlink noise. We have assumed it to be independent of the direction of propagation of the optical beams within. The vector $\mathbf{\delta}_{ji}$ is the test-mass displacement in metres, including displacement produced by acceleration noise; it is not itself an acceleration. With the same sampling convention, the reference and test-mass clock coefficient is

$$\beta_{ji} = \nu_{ji} - \nu_{jk}. \quad (12)$$

It has the opposite sign to the adjacent-minus-local beat frequency, which explains the plus sign multiplying $\beta_{ji} q_j$ in the readouts.

These equations retain the terms needed to explain laser- and clock-noise suppression. A full instrument model includes further effects, such as tilt-to-length coupling and additional stray-light and optical-path disturbances. The miniLISA model omits the test-mass and reference interferometers and their associated bench-motion, backlink, and test-mass noise terms.

B. Laser Noise and TDI

In the LISA measurement system, laser frequency noise is first reduced onboard by locking one laser to an optical resonator and phase-locking the remaining lasers to this pre-stabilized reference with fixed frequency offsets. The last step ensures that all heterodyne beat notes remain within the phasemeter bandwidth while maintaining coherent phase relationships between spacecraft. However, residual laser frequency noise remains many orders of magnitude above the displacement sensitivity requirement and must be suppressed in post-processing. TDI achieves this by forming linear combinations of one-way phase measurements and appropriately delayed copies thereof, effectively synthesizing an equal-arm interferometer in post-processing.

For scale, consider a representative pre-stabilized frequency-noise amplitude of $30 \text{ Hz}/\sqrt{\text{Hz}}$ [3]. At Fourier frequency $f = 10 \text{ mHz}$, the corresponding single-laser phase-noise amplitude is $30/(2\pi f) \simeq 480 \text{ cycles}/\sqrt{\text{Hz}}$, equivalent to about $0.5 \text{ mm}/\sqrt{\text{Hz}}$ displacement noise at 1064 nm . This illustrative conversion is many orders of magnitude above picometre-scale displacement noise required for GW detection.

The construction proceeds in two steps: local measurements first remove optical-bench motion and reduce the six laser noises to three; delayed combinations of the resulting links then suppress the remaining laser noise [12, 13]. Using Eqs. (7), (10), and (11), we form

$$\begin{aligned} T_{ji} &= \text{tm}_{ji} - \text{ref}_{ji}, \\ \xi_{ji} &= \text{sci}_{ji}^c - \frac{1}{2} (T_{ji} + \mathbf{D}_{ji} T_{ij}). \end{aligned}$$

Within the common-optical-frequency approximation above, the bench terms cancel, leaving the test-mass contribution

$$M_{ji} = -\frac{\nu_0}{c} [\mathbf{n}_{ji} \cdot \boldsymbol{\delta}_{ji}(t) + \mathbf{D}_{ji}(\mathbf{n}_{ij} \cdot \boldsymbol{\delta}_{ij}(t))].$$

The following equations omit readout noises. Reciprocal backlink noise cancels in the local reference difference

$$z_{ji} = \frac{1}{2} (\text{ref}_{ji} - \text{ref}_{jk}) = \delta\phi_{jk} - \delta\phi_{ji} + \beta_{ji} q_j.$$

For cyclic triples $(i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$, the intermediate variables can then be defined as

$$\begin{aligned} \eta_{ji} &= \xi_{ji} - z_{ji} \\ &= \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{jk}(t) - (\alpha_{ji} + \beta_{ji}) q_j(t) \\ &\quad + \delta\phi_{ji}^{\text{GW}}(t) + M_{ji}(t), \\ \eta_{ij} &= \xi_{ij} + \mathbf{D}_{ij} z_{ji} \\ &= \mathbf{D}_{ij} \delta\phi_{jk}(t) - \delta\phi_{ij}(t) - \alpha_{ij} q_i(t) \\ &\quad + \mathbf{D}_{ij} [\beta_{ji} q_j(t)] \\ &\quad + \delta\phi_{ij}^{\text{GW}}(t) + M_{ij}(t). \end{aligned}$$

Here M_{ij} is obtained by exchanging i and j in M_{ji} . These combinations reduce the number of independent laser phase noises from six to three, retaining $\delta\phi_{12}$, $\delta\phi_{23}$ and $\delta\phi_{31}$, while removing the optical-bench displacement terms. The construction introduces an asymmetry between the two link orientations.

For the two-arm geometry used below, we use the Michelson observable X_i . Define the synthesized round-trip measurements

$$R_{ij} = \eta_{ij} + \mathcal{D}_{ij} \eta_{ji}, \quad R_{ik} = \eta_{ik} + \mathcal{D}_{ik} \eta_{ki}.$$

The first-generation combination is

$$X_i = (1 - \mathcal{D}_{ij} \mathcal{D}_{ji}) R_{ik} - (1 - \mathcal{D}_{ik} \mathcal{D}_{ki}) R_{ij}. \quad (13)$$

Here $\mathbf{D}$ describes propagation in the measurement chain, while $\mathcal{D}$ denotes a delay applied in post-processing. Cancellation requires the processing delays to match the physical delays. A mismatch δd leaves a phase residual proportional to $\delta d \delta\phi$, so either an inaccurate delay estimate or an untracked change in the physical delay can limit laser-noise suppression.

First-generation TDI combinations cancel laser noise for constant light-travel times. Orbital motion makes

these delays time-dependent, so their order of application matters and the cancellation is no longer exact. Second-generation combinations use longer sequences of delays to cancel the leading residual from arm-length changes, leaving terms involving higher powers of the arm velocities and arm accelerations [13].

C. Clock-noise Coupling and Correction

Clock timing jitter is another primary noise in the raw measurements [2, 14]. Similarly to laser frequency noise, it exceeds the sensitivity requirement by several orders of magnitude and must be removed in post-processing before gravitational wave signals can be recovered. Unlike laser frequency noise, it couples into the measurement only locally at one spacecraft. Hence, to still allow for a similar differential measurement of the clock noise, the laser fields exchanged between the satellites are modulated with a signal derived from the USOs. From the upper-sideband and carrier phase measurements at the science interferometer, a correcting term can be constructed.

Using the phase-domain counterpart of the clock-transfer construction in [14], and neglecting readout and modulation noises initially, we define

$$r_{ji} = \frac{\text{sci}_{ji}^{\text{sb},+}(t) - \text{sci}_{ji}^c(t)}{\nu_{ij}^m} = \mathbf{D}_{ji} q_i(t) - q_j(t). \quad (14)$$

The retained GW phase cancels in this difference because we have approximated the carrier and sideband GW couplings as equal. Keeping their optical-frequency difference would instead add $-H_{ji}$ to r_{ji} ; the cancellation is therefore approximate. The normalization uses the transmitted modulation frequency ν_{ij}^m , giving r_{ji} units of seconds. Equivalently, $r_{ji} = (\text{sci}_{ji}^{\text{sb},+} - \text{sci}_{ji}^{\text{sb},-}) / (2\nu_{ij}^m)$ under the same approximations. These variables contain delayed remote clock noise minus local clock noise, matching the structure of laser noise in the science readouts. Delayed combinations of the r variables can therefore be used to construct clock corrections for the TDI observables [14]. The modulation tone must carry the same timing fluctuations that enter the phase measurement. In LISA, pilot-tone correction removes the ADC sampling jitter, making the pilot tone the relevant clock reference. Noise added between this reference and the modulation tone limits the subsequent clock correction. Let $\tau_{ij}^{\text{mod}}(t)$ denote differential timing noise in the tone derivation for MOSA ij , in seconds, and let $\psi_{ij}^{\text{mod}}(t)$ denote additional optical modulation phase noise, in cycles. The excess upper-sideband phase is

$$m_{ij}(t) = \nu_{ij}^m \tau_{ij}^{\text{mod}}(t) + \psi_{ij}^{\text{mod}}(t).$$

Including these terms and readout noise gives

$$r_{ji} = \mathbf{D}_{ji}q_i - q_j + \frac{\mathbf{D}_{ji}m_{ij} - m_{ji}}{\nu_{ij}^m} + \frac{N_{ji}^{\text{sb},+} - N_{ji}^c}{\nu_{ij}^m}.$$

Modulation and readout noise are kept separate because their transfer functions differ. A source modulation error appears in both local and delayed measurements, while a readout error is added in a particular measurement channel. Independent sources may be uncorrelated, but repeated appearances of the same source must still be combined before calculating its output noise power.

Dividing the modulation phase by its frequency gives $m_{ij}/\nu_{ij}^m = \tau_{ij}^{\text{mod}} + \psi_{ij}^{\text{mod}}/\nu_{ij}^m$. A higher modulation frequency therefore reduces the effect of a fixed phase noise, but leaves a fixed timing noise unchanged. Improving that timing-noise floor requires a quieter tone-generation chain. When the local and transmitted frequencies differ, the local timing contribution retains the factor ν_{ji}^m/ν_{ij}^m . A measurement at one modulation frequency cannot distinguish the two contributions.

The two modulation-frequency synthesis chains can add different excess noises. Local reference-interferometer sideband measurements provide additional observables for correcting the accessible differential part [14]. This contribution is absent from the present one-modulator-per-spacecraft testbed.

III. MINILISA DESIGN

A. Motivation for the Present Work

The present work extends the electronic delay-line testbed with optical inputs and independent spacecraft clocks. It combines physical laser noise, sideband-based clock transfer, and hardware delays of several seconds, and uses component measurements to estimate the performance of the combined setup.

The electronic delays place samples acquired several seconds apart in the same interferometric measurement. This lets us test noise cancellation with independently clocked hardware, including changes in the delay caused by the hardware clock itself.

B. Measurement Philosophy

The design provides six directed links between three spacecraft, allowing LISA TDI combinations to be formed within the measurement model described below. We use the familiar Michelson X for the performance analysis; it uses four of these links. The testbed reproduces the laser frequency noise (LFN), clock-noise, and GW couplings of the LISA measurements. The first index of a link denotes the receiving spacecraft (SC); source

quantities carry a single spacecraft index, $\delta\phi_i$, q_i , and ν_i^m . The target carrier phase fluctuations are

$$\delta\phi_{ij}^c(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t),$$

and the upper-sideband fluctuations are

$$\delta\phi_{ij}^{\text{sb},+}(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - (\alpha_{ij} + \nu_j^m)q_i(t) + \nu_j^m\mathbf{D}_{ij}q_j(t).$$

Here α_{ij} is the signed nominal carrier beat frequency. The sideband sampling coefficient is $\gamma_{ij} = \alpha_{ij} + \nu_j^m - \nu_i^m$; combining it with the local modulation phase gives the clock term above. Both channels use the common GW phase approximation of Section II. We call a measurement *LISA-like* when it reproduces these couplings. Modulation, board, and readout noises are added below.

An FPGA delay line supplies the propagation delay and the prescribed Doppler and GW frequency shifts. Heterodyning the optical fields with a common reference laser brings their phase information into the MHz range accessible to the electronics. The reference laser noise becomes part of the effective input laser noise, whose cancellation is discussed below.

The testbed omits the test-mass and reference interferometers, together with test-mass acceleration noise, optical-bench motion, backlink noise, and tilt-to-length coupling. Consequently, the local combinations T_{ji} and ξ_{ji} used to remove bench motion in Section II are unnecessary. Identifying the two bench lasers on each spacecraft with a single source also removes the need for the reference differences z_{ji} . The synthesized links reproduce the three-laser structure of the η observables, but omit the reference-interferometer clock terms proportional to β_{ji} . The clock correction tested here is therefore the reduced case of the LISA construction with those terms absent.

C. Optical Input

Three lasers represent the spacecraft, each with an independent clock driving its modulation and phase measurement. A fourth laser provides the common optical reference. The role of the reference laser is to convert the optical phase fluctuations of each SC laser into electronically processable heterodyne beat notes, which can then be measured by the delay line. This heterodyning step effectively downconverts the optical signals from approximately 281 THz to tens of megahertz, preserving laser frequency noise in the millihertz range of interest. Among the SC lasers, one is frequency-stabilized to a 25 cm ultra-low expansion (ULE) glass cavity, while the reference laser is locked to a 30 cm ULE cavity via the Pound-Drever-Hall method. Both cavities are housed in a vacuum chamber surrounded by three layers of aluminum thermal shielding. The target is a pre-stabilized LFN level of around 30 Hz/ $\sqrt{\text{Hz}}$ in the millihertz band.

The remaining two SC lasers are PLL-locked to the synthetic LISA-like beat notes.

Each spacecraft uses an independent rubidium frequency standard as the Moku reference. The Moku supplies both the phase readout and the modulation tone driving the fiber-coupled EOM. We use q_i for the timing fluctuations common to the EOM drive and the phasemeter's sampling reference. Excess noise between these paths enters the modulation and readout terms separately.

The reference laser is unmodulated in the configuration described here, so the spacecraft carrier and sidebands each beat with the reference carrier. The modulation frequency must therefore keep all three electrical beat notes within the delay-line input bandwidth. These input tones carry the optical phase and clock information; the LISA-like carrier-carrier and sideband-sideband readouts are formed by digital subtraction downstream. Modulating the reference laser is a possible extension for higher modulation frequencies, discussed in Section V C. The construction of a two-way arm link is illustrated in the lower panel of Fig. 1, alongside a simplified schematic of the corresponding LISA link showing only the functionally relevant elements. Taking the signed photodetector beat phase as SC laser minus reference laser, the carrier signal phase before additive detector noise is

$$\Phi_{PD,i}^c(t) = \nu_{R,i}t + \delta\phi_i^{\text{eff}}(t),$$

where $\nu_{R,i}$ is the signed nominal beat note frequency between the reference laser and the SC i laser, and $\delta\phi_i^{\text{eff}} = \delta\phi_i - \delta\phi_R$ includes their relative laser phase noise. The upper-sideband phase is

$$\begin{aligned} \Phi_{PD,i}^{\text{sb},+}(t) = & (\nu_{R,i} + \nu_i^m)t \\ & + \delta\phi_i^{\text{eff}}(t) + \nu_i^m q_i(t) \\ & + \nu_i^m \tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t). \end{aligned}$$

Here, τ_i^{mod} is timing noise added by the modulation path relative to the common reference q_i , while ψ_i^{mod} is additional modulation phase noise. Noise added only by the sampling path is included in the readout terms below. These are the single-index counterparts of τ_{ij}^{mod} and ψ_{ij}^{mod} in Section II C, with the same dependence on modulation frequency. We write their combined upper-sideband phase noise as

$$m_i(t) = \nu_i^m \tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t).$$

The lower sideband carries the opposite modulation phase, $-\nu_i^m q_i - m_i$. If its signed electrical frequency $\nu_{R,i} - \nu_i^m$ is negative, representing it at positive frequency also reverses the measured phase sign; the clock correction below uses the upper sideband.

The photodetector output is then split and distributed across three delay-line boards. One of these boards synthesizes the local LISA-like beat note signals, the other two do so for the other two SC. The signal routing between the boards is illustrated in Fig. 2.

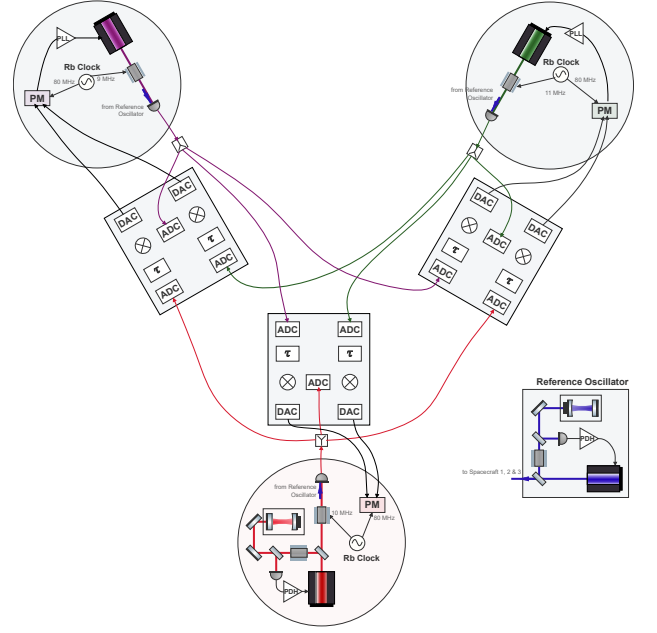


FIG. 2. Planned six-link routing architecture for miniLISA. Each spacecraft has an optical source and an Rb-referenced Moku phasemeter that also supplies its EOM drive; each delay-line board can synthesize two incoming links. The complete routing supports six directed links, whereas the Michelson calculation in Section IV uses links 12, 21, 13 and 31. The spacecraft-1 and reference lasers are stabilized to separate optical cavities; the other spacecraft lasers are phase-locked to synthesized beat notes. The common reference laser supplies the optical heterodyne reference.

Draft note: Update the schematic labels from the legacy modulation frequencies to (35, 36, 34) MHz and identify the independent Rb-referenced Moku chains.

Shot noise, intensity noise and detector electronics add fluctuations to the photocurrent, which produce phase errors when the delay-line PLLs track the beat notes. Noise added before the electrical split is shared between boards; ADC and electronics noise added after the split need separate terms. Their phase-noise scales are calculated in Section IV B, and their different propagation is described in Section IV C.

D. The Delay Line

The digital delay line emulates the light-travel time along each directed link. The current firmware always tracks and stores the carrier and both sidebands. At the resulting data rate, the available memory supports a maximum delay of approximately 8.5 s, covering LISA's nominal one-way light-travel time of about 8.3 s.

The implementation uses an AMD Xilinx ZCU208 RF-SoC evaluation board, with integrated ADCs and DACs, FPGA programmable logic, and an embedded processing unit (PS). For each synthesized link, separate DPLLs

track the carrier and sidebands of the remote and local input signals. The tracked remote frequencies are buffered to impose the arm delay, then combined with the current local frequencies. The present 125 MHz input-processing rate restricts the optical heterodyne tones to below 62.5 MHz. This is a constraint on the laboratory optical frequency plan, distinct from the frequencies of the synthesized LISA-like outputs.

Orbital delays and GW link responses are computed in advance in Python and uploaded to the PS. During playback, the PS converts the model inputs into fixed-point commands and streams timestamped records to the programmable logic. The buffer supplies the macroscopic light-travel time, while fine delay actuation follows the prescribed arm-length evolution using fractional-delay interpolation. The underlying electronic architecture is described in [9].

The FPGA subtracts the local frequency from the delayed remote frequency and adds the prescribed Doppler and GW offsets. Integrating this frequency produces the output phase for each of the carrier and sideband tones, which are combined at the DAC. Section III E describes the GW scaling. Omitting the nominal frequency evolution, the intended carrier phase is

$$\mathbf{D}_{ij}\delta\phi_j^{\text{eff}} - \delta\phi_i^{\text{eff}} + \delta\phi_{ij}^{\text{GW}}.$$

The performance calculation below uses static delays.

The two ADC inputs and the DAC output share the timing jitter $\epsilon_i(t)$ of the board on spacecraft i . Sampling imprints this jitter on both inputs. The remote contribution then passes through the delay, while subtracting the local signal in the digital mixer reverses the sign of its ADC contribution. The DAC adds jitter at the output beat frequency α_{ij} , giving

$$\begin{aligned}\delta\phi_{\epsilon,ij}^c(t) &= -\nu_{R,j}\mathbf{D}_{ij}\epsilon_i(t) \\ &\quad + \nu_{R,i}\epsilon_i(t) + \alpha_{ij}\epsilon_i(t) \\ &= \nu_{R,j}(1 - \mathbf{D}_{ij})\epsilon_i(t).\end{aligned}$$

Here $\alpha_{ij} = \nu_{R,j} - \nu_{R,i}$, so the local ADC and DAC terms combine to give $\nu_{R,j}\epsilon_i(t)$.

The same cancellation applies to the upper sideband, with the remote input frequency replaced by $\nu_{R,j} + \nu_j^m$:

$$\delta\phi_{\epsilon,ij}^{\text{sb},+}(t) = (\nu_{R,j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t).$$

Both expressions assume constant nominal frequencies and no additional output frequency offset.

The same board jitter also changes the physical duration of a programmed delay. For a constant command d_{ij} in board time $u_i(t) = t + \epsilon_i(t)$, the input and output times satisfy $u_i(t) - u_i(t_{\text{in}}) = d_{ij}$. To first order,

$$d_{ij}^{\text{phys}}(t) = t - t_{\text{in}} = d_{ij} + \delta d_{ij}^{\epsilon}(t), \quad (15)$$

$$\delta d_{ij}^{\epsilon}(t) = -(1 - \mathbf{D}_{ij})\epsilon_i(t). \quad (16)$$

Thus the physical delay can fluctuate even when its programmed value is known exactly. In LISA, ranging errors leave uncertainty in the light-travel times used for

TDI. Here, board timing changes the realized delay itself. Both effects can leave laser noise if the delays applied in processing do not match the physical delays.

E. GW Signal Injection

The GW phase defined in Eq. (4) is implemented electronically after the background delay. The required phase on the link from j to i is

$$\delta\phi_{ij}^{\text{GW}}(t) = -\nu_j H_{ij}(t).$$

Since the nominal optical frequency ν_j is constant, the frequency offset supplied to the phase accumulator is

$$\delta\nu_{ij}^{\text{inj}}(t) = \delta\dot{\phi}_{ij}^{\text{GW}}(t) = -\nu_j \dot{H}_{ij}(t). \quad (17)$$

Accumulating this offset reproduces the required GW phase up to an initial phase constant. The scale factor is the represented optical frequency, not the MHz beat note frequency entering the delay-line ADC. Thus the GW signal is generated by adding phase evolution to the reconstructed signal; the hardware delay itself remains d_{ij} and is not additionally perturbed by H_{ij} . This implements the first-order propagation model of Section II.

Each directed-link waveform is evaluated at its reception time t . The propagation and source geometry are already included in H_{ij} , so no additional background delay is applied to the injection waveform after it is generated. Relative timing between the playback streams must reproduce this convention.

The same GW offset is applied to the carrier and both sidebands, following the approximation in Section II. It therefore cancels from their difference used for clock correction while remaining in the science signal.

When the link response is supplied as a fractional-frequency signal $y_{ij} = -\dot{H}_{ij}$, the command is $\delta\nu_{ij}^{\text{inj}} = \nu_j y_{ij}$, using $\nu_j \simeq c/(1064 \text{ nm})$ for the optical scale. The link responses can be computed in Python for a chosen waveform and constellation geometry. Multiple sources can be combined by summing their link responses in the linear regime. The source class is not fixed by the hardware, but the resulting commands must lie within its frequency range, resolution and update bandwidth.

A signed 32-bit GW command is added on the 64-bit phase-increment scale. At a 125 MHz accumulator clock, one least significant bit corresponds to $125 \text{ MHz}/2^{64} \simeq 6.78 \times 10^{-12} \text{ Hz}$.

Draft note: Generate the delay commands and GW responses from the same orbit and reception-time convention. State the command update rate and verify the recovered amplitude and phase, including quantization and playback timing.

F. Readout and Data Processing

The electrical signals produced by the delay line are digitized and tracked by Moku phasemeters, with each

spacecraft's phase measurements referenced to its own clock.

The synthesized links can be used directly as η observables. Adding in the modulation, board, and readout noises, the carrier measurement is

$$\eta_{ij}(t) = \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t) + \nu_{R,j}(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^c(t),$$

and the corresponding upper-sideband–upper-sideband beat note as

$$\begin{aligned} \eta_{ij}^{\text{sb},+}(t) = & \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \gamma_{ij}q_i(t) \\ & + \nu_j^m \mathbf{D}_{ij}q_j(t) - \nu_i^m q_i(t) \\ & + \mathbf{D}_{ij}[\nu_j^m \tau_j^{\text{mod}}(t) + \psi_j^{\text{mod}}(t)] \\ & - \nu_i^m \tau_i^{\text{mod}}(t) - \psi_i^{\text{mod}}(t) \\ & + (\nu_{R,j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^{\text{sb},+}(t). \end{aligned}$$

The terms proportional to q_i describe timing common to the modulation and phasemeter reference, while those proportional to ϵ_i come from the delay-line board.

The additive phase noise b_{ij}^a combines optical readout, residual board, and final-phasemeter noise, in cycles; timing and modulation noise are included explicitly. To retain the noise shared by copies of a photodetector signal, write

$$b_{ij}^a = \mathbf{D}_{ij}p_j^a - p_i^a + e_{ij}^a, \quad (18)$$

where p_i^a is the equivalent input phase noise in channel a before the split, and e_{ij}^a contains noise added by the individual electronic paths. This expression assumes matched tracking responses. Any mismatch must be included in the transfer functions rather than treating copies of p_i^a as independent sources.

The effective laser phases $\delta\phi_i^{\text{eff}}$ enter with the same delayed-difference structure as the LISA laser phases. TDI therefore suppresses both the spacecraft and reference laser noise.

Finally, the upper-sideband-minus-carrier variable takes the form

$$\begin{aligned} r_{ij} = & \frac{\eta_{ij}^{\text{sb},+} - \eta_{ij}}{\nu_j^m} \\ = & \mathbf{D}_{ij}q_j - q_i + \frac{\mathbf{D}_{ij}m_j - m_i}{\nu_j^m} \\ & + (1 - \mathbf{D}_{ij})\epsilon_i + \frac{b_{ij}^{\text{sb},+} - b_{ij}^c}{\nu_j^m}. \end{aligned}$$

The common GW phase cancels, leaving the clock-transfer signal together with modulation, board and readout noise.

The following common-mode cancellation assumes a shared board timing reference, with residual differences between distribution paths. Writing $\epsilon_i = \epsilon_{\text{com}} + \Delta\epsilon_i$ separates their shared fluctuation from the differential jitter. In the constant-frequency model above, the common

term can be absorbed into the effective laser phases and clock errors as $\delta\phi_i^{\text{eff}} - \nu_{R,i}\epsilon_{\text{com}}$ and $q_i - \epsilon_{\text{com}}$. It therefore cancels under TDI together with clock correction. Differential board jitter remains and sets the relevant timing residual.

Draft note: State which oscillator references the delay-board ADCs, DACs and delay counters, and how it is distributed. Confirm whether the boards share one reference or follow their spacecraft Rb clocks. In the latter case, specify the correlations between ϵ_i and q_i and revise the timing budget accordingly.

Table I summarizes the shared and laboratory-specific noise sources. The readout predictions are derived in Section IV B.

TABLE I. Noise sources shared with LISA or introduced by the miniLISA implementation, and their treatment in the present static analysis. Sources omitted from the testbed are described in Section III.

Noise source	LISA	miniLISA	Treatment / relevance
Present in both LISA and miniLISA			
Laser frequency noise	Present	Present	Suppressed by TDI with matched delays; ideal cancellation in this estimate.
Measurement-clock jitter q_i	Present	Present	Sideband-based post-processing correction; residual tone-generation noise remains.
Modulation noise	Present	Present	Limits clock correction. Estimator assigned to source noise; channel-noise alternative remains to be tested.
Science readout noise N_{ij}^c, N_{ij}^{sb}	Weak received light; shot noise important	Bright beams; shared detector inputs	Shared carrier noise cancels in ideal TDI; channel differences enter clock correction.
ADC and phasemeter noise	Present	Present	Ideal ADC quantization included; sampling jitter and residual electronics treated separately.
Delay mismatch / processing residuals	Ranging and processing errors	Board-induced delay fluctuations and processing errors	Track effective delays; only matched static delays are used in the present estimate.
Additional sources in miniLISA			
Shared reference-laser noise	No laboratory reference laser	Present	Part of the effective source phase; suppressed by TDI in the ideal model.
Delay-board timing jitter ϵ_i	No electronic arm-delay board	Present	Shared jitter cancels in TDI with clock correction; differential jitter can remain.
Residual delay-board electronics	No electronic arm-delay board	Present	Measured baseline included as additive noise; not assumed to cancel.

IV. PERFORMANCE MODELLING

We estimate the noise remaining after TDI and clock correction from the measured delay-line noise, the modulation characterization and a prospective optical readout calculation. The calculation uses static delays and ideal cancellation of the common laser and clock noises. It is a conditional component-noise estimate, not a measured sensitivity of the assembled optical testbed; delay-knowledge and interpolation residuals are not included.

A. Noise Inputs and Comparison Reference

For laser frequency noise we adopt the LISA pre-stabilization reference [3],

$$A_\nu(f) = 30 \frac{\text{Hz}}{\sqrt{\text{Hz}}} \sqrt{1 + \left(\frac{2 \text{ mHz}}{f}\right)^4}. \quad (19)$$

Its equivalent phase ASD is $A_\phi = A_\nu/(2\pi f)$.

The clock estimate uses the phase-noise model of a 10 MHz Rb standard. Its single-sideband phase noise is anchored at $\mathcal{L}(10 \text{ Hz}) = -130 \text{ dBc/Hz}$, consistent with the specifications in [15]. Below this point, the phase PSD is extrapolated as f^{-1} down to 1 Hz, f^{-2} down to 10 mHz, and f^{-3} below 10 mHz. The corresponding

timing ASD is

$$A_q(f) = \frac{\sqrt{2 \cdot 10^{\mathcal{L}(f)/10}}}{2\pi \times 10 \text{ MHz}}. \quad (20)$$

This gives approximately 2.25 ps/ $\sqrt{\text{Hz}}$ at 10 mHz and 71 ps/ $\sqrt{\text{Hz}}$ at 1 mHz. It is an estimate from the assumed low-frequency continuation, rather than a measurement of the laboratory clocks. The common timing noise couples to a carrier link as $|\alpha_{ij}|A_q$ and is removed by the ideal clock correction used here. At $|\alpha| = 6.6 \text{ MHz}$ the assumed raw single-link clock coupling is only 1.49×10^{-5} cycles/ $\sqrt{\text{Hz}}$ at 10 mHz, so a measurable suppression margin cannot be assumed across the band.

Draft note: Measure the timing noise of the actual Rb-referenced Moku chain and plot the uncorrected clock contribution through the science TDI weights, summing each shared clock coherently across links. If it is too close to the secondary floor, use a calibrated clock perturbation or a revised beat-frequency plan and state the resulting suppression-test band.

The electronic input is the measured delay-line phase-noise floor, including processing and electronics and possibly the readout phasemeter's ADC. The carrier and sideband measurements give the same floor, so we assign the same PSD to each electronic readout term. The terms are taken to be independent between links and channels, while repeated uses of a given readout remain coherent.

The modulation characterization uses half the upper-minus-lower sideband phase of a local optical beat, with

a common modulation and measurement reference. For the plotted budget, we assign this spectrum to independent source modulation phases m_i . This is an assumption about its origin: the estimator also contains additive channel noise and does not establish the correlations of a physical source shared between links.

Draft note: Separate source modulation noise from additive measurement noise, for example using simultaneous channel cross-spectra. Recalculate the budget with the estimator assigned to additive channel terms as an alternative scenario; these two assignments are not guaranteed upper and lower bounds.

Figure 3 shows these component inputs in cycles/ $\sqrt{\text{Hz}}$. The measured spectra cover the plotted 0.25 mHz–1 Hz band without extrapolation.

For comparison, we use the single-link reference adopted in the component analysis. With $\lambda = 1064 \text{ nm}$, its equivalent phase ASDs are

$$A_{\text{OMS}}(f) = \frac{15 \times 10^{-12} \text{ m}}{\lambda} \sqrt{1 + \left(\frac{2 \text{ mHz}}{f}\right)^4}, \quad (21)$$

$$A_{\text{acc}}(f) = \frac{3 \times 10^{-15} \text{ m s}^{-2}}{(2\pi f)^2 \lambda} \sqrt{1 + \left(\frac{0.4 \text{ mHz}}{f}\right)^2} \times \sqrt{1 + \left(\frac{f}{8 \text{ mHz}}\right)^4}. \quad (22)$$

The reference is $A_{\text{link}} = (A_{\text{OMS}}^2 + A_{\text{acc}}^2)^{1/2}$. Acceleration noise is included here as a comparison quantity; it is not an experimentally present test-mass noise in the current testbed. The component spectra are displayed directly, without applying a TDI transfer function. This figure therefore compares input noise scales.

B. Predicted Carrier and Sideband Readout Noise

For the readout calculation, we consider one phase-modulated laser interfering with an unmodulated reference laser in a prospective Newport PIN-detector readout. Each beam delivers 4 mW at the detector, after any combining losses, and the phase-modulation depth is $\mu_{\text{EOM}} = 0.53 \text{ rad}$. These numerical values apply to carrier–carrier and sideband–carrier detection in this configuration.

With responsivity 0.6 A/W and heterodyne power-overlap efficiency 0.9, the carrier and first-order sideband peak beat currents are 4.24 and 1.16 mA. Including shot noise, dark-current shot noise, Johnson noise, additive RF intensity noise and ADC quantization gives phase ASDs of 3.40×10^{-9} and 1.24×10^{-8} cycles/ $\sqrt{\text{Hz}}$, respectively. The receiver parameters are recorded in the caption of Fig. 3; the dark current is 100 nA and the gain after the ideal three-way split is $50/\sqrt{3} \text{ V/A}$. The calculation assumes locally white additive noise around each RF beat and combines both RF noise sidebands in demodulation.

The bright laboratory beams make this contribution small compared with the measured electronic and sideband-difference spectra. Detector noise shared before the electrical split has the laser-noise structure: its carrier contribution cancels in ideal TDI, while the carrier–sideband difference χ_i enters clock correction as derived below. Noise added independently after the split uses the separate link transfer functions.

C. Propagation Through TDI and Clock Correction

The propagation calculation uses the four links of Michelson X , static delays and constant nominal frequencies. The illustrative frequency plan is $\nu_{\text{R},i} = (15.0, 10.9, 21.6) \text{ MHz}$ and $\nu_i^m = (35, 36, 34) \text{ MHz}$, giving $(\alpha_{12}, \alpha_{21}, \alpha_{13}, \alpha_{31}) = (-4.1, 4.1, 6.6, -6.6) \text{ MHz}$. The nominal one-way delay is $L = 2.5 \times 10^9 \text{ m/c} \simeq 8.339 \text{ s}$. These are example operating frequencies. The small modulation-frequency offsets separate the synthesized carrier and sideband beats; the laboratory readout is not constrained to the representative LISA 5–25 MHz band. Writing $z_{ij} = \exp(-2\pi i f d_{ij})$, $A = z_{12} z_{21}$ and $B = z_{13} z_{31}$, the static Michelson defined above becomes

$$X = (1 - A)(\eta_{13} + z_{13}\eta_{31}) - (1 - B)(\eta_{12} + z_{12}\eta_{21}). \quad (23)$$

Let P_{ij} denote its four link coefficients and K_{ij} the clock-correction coefficients, chosen so that

$$X_c = \sum_{ij} (P_{ij}\eta_{ij} + K_{ij}r_{ij}) \quad (24)$$

has zero coefficient for each q_i . The sum runs over links 12, 21, 13 and 31, using the upper-sideband-minus-carrier definition of r_{ij} given above. The static second-generation convention used in the calculation is $X_{2c} = (1 - AB)X_c$; this extra filter must multiply both the science and correction weights. Laser and clock cancellation can be checked algebraically for unequal static delays. The calculation is restricted to static arms.

For noise added to individual links, repeated appearances of one readout must be combined coherently. In particular, for the first-generation weights the coefficients are

$$T_{b_{ij}^c} = P_{ij} - \frac{K_{ij}}{\nu_j^m}, \quad T_{b_{ij}^{\text{sb},+}} = \frac{K_{ij}}{\nu_j^m}. \quad (25)$$

The two appearances of a given carrier noise are therefore combined before taking a squared magnitude. These coefficients apply directly to e_{ij}^a in Eq. (18), not to independent copies of a shared detector noise.

For matched paths, the shared carrier term $\mathbf{D}_{ij}p_j^c - p_i^c$ has the laser-noise structure and cancels in carrier TDI. However, define the detector channel difference $\chi_i =$

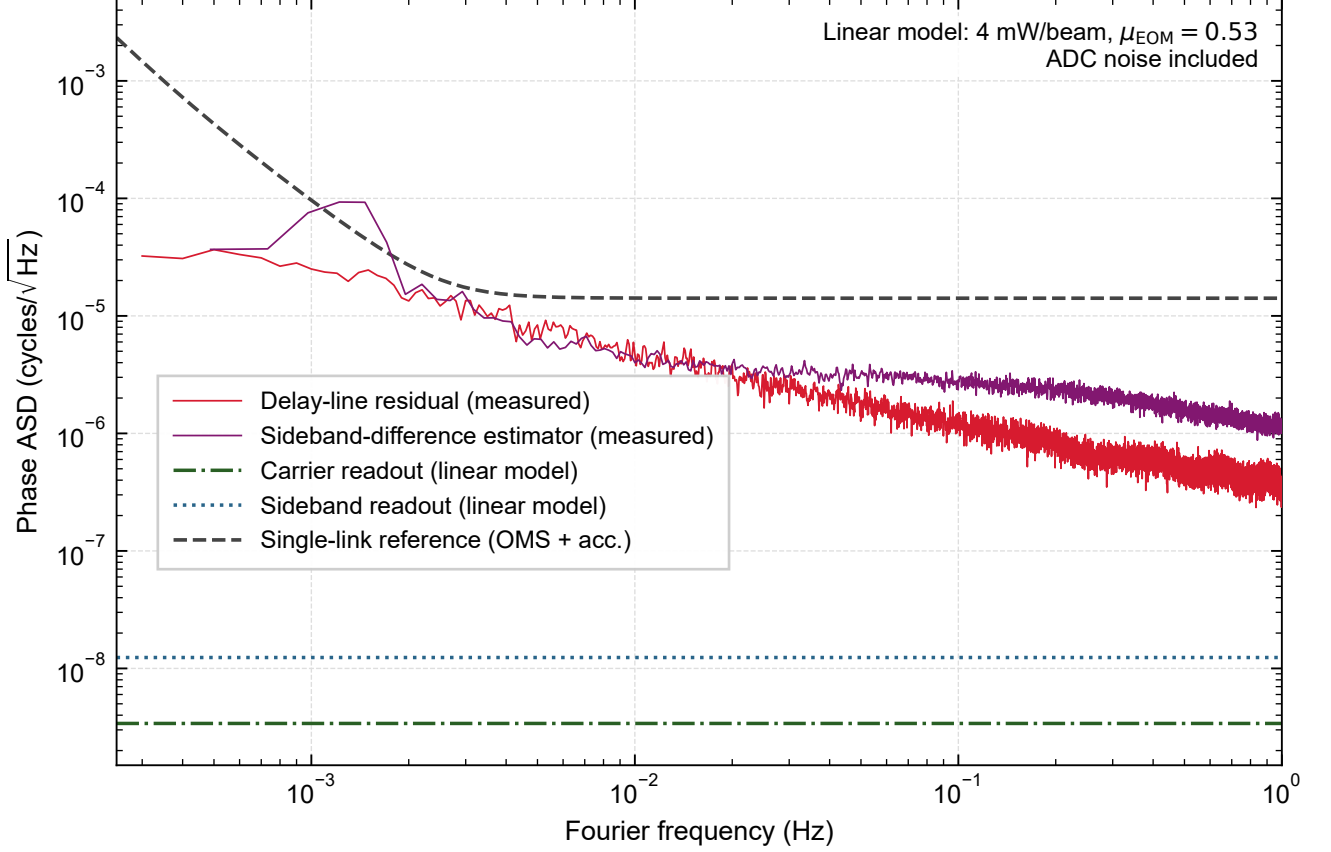


FIG. 3. Component phase-noise amplitude spectral densities compared with the single-link reference formed from optical metrology and acceleration noise. Solid curves show the measured delay-line residual and modulation phase-noise estimator; measured Fourier bins are plotted without smoothing. The illustrative linear-receiver carrier and first-order sideband readout noises include shot noise, dark-current shot noise, Johnson noise, relative intensity noise and ADC quantization noise. The optical model assumes one phase-modulated beam and one unmodulated reference beam, each delivering 4 mW at the detector, with modulation depth 0.53 rad. The detector responsivity is 0.6 A/W, heterodyne efficiency is 0.9, and the termination is 50 Ω at 300 K. The ADC model uses a 1 V full-scale span, 14 bits and 2 GS/s, with an ideal three-way electrical split. The assumed white RIN is $0.9 \times 10^{-8}/\sqrt{\text{Hz}}$, summed coherently between the beams. The model uses (35, 36, 34) MHz modulation and electrical carrier frequencies of 15.0, 10.9 and 21.6 MHz; the readout prediction assumes white noise across the corresponding RF bands. All curves are shown before propagation through TDI and clock correction; the measured electronic residual is shown as characterized and is not deconvolved into individual board noise sources. The assumptions of the component budget are stated in Section IV.

$p_i^{\text{sb},+} - p_i^c$. It contributes to the clock-transfer variable as

$$r_{ij}^{\text{det}} = \frac{z_{ij}\chi_j - \chi_i}{\nu_j^m}, \quad (26)$$

and hence to the corrected observable as

$$X_{2c}^{\text{det}} = (1 - AB) \sum_{ij} K_{ij} \frac{z_{ij}\chi_j - \chi_i}{\nu_j^m}. \quad (27)$$

For the estimate, detector noise in the carrier and sideband channels is taken to be independent, so $S_{\chi_i} = S_{p_i^{\text{sb},+}} + S_{p_i^c}$. Copies of each detector channel shared between boards retain the same noise realization.

For independent physical sources n_a , the output PSD is

$$S_{X_{2c}}(f) = \sum_a |T_a(f)|^2 S_a(f). \quad (28)$$

The measured electronic spectrum is propagated with the carrier and sideband coefficients above. The predicted detector terms use Eq. (27). ADC quantization is shown in the input readout estimate but is not added separately to the total, where electronics are represented by the measured delay-line floor.

The modulation-frequency scaling depends on whether the input is direct phase noise or timing noise, as described in Section II C; it cannot be inferred from the

full sideband-difference estimator alone. The TDI differencing response also acts on a GW signal, so a small output ASD alone does not imply improved sensitivity.

D. Estimated Residual Noise

Figure 4 combines the electronic, modulation and detector contributions after static second-generation TDI and clock correction. Each physical source is multiplied by its transfer function before the output PSDs are summed. The calculation uses four independent electronic link readouts per channel, three shared detector inputs, and the source-modulation assignment discussed above. The detector contribution is small compared with the measured electronic and modulation inputs, consistent with the use of bright laboratory beams rather than a weak-light readout.

The nominal beat-frequency coupling of board timing is already included in the ϵ_i terms and must not be added again as independent ranging noise. Differential timing and imperfect delay knowledge remain unquantified in this budget.

Draft note: Add a laser-noise residual using measured or bounded errors in the realized and processing delays, including interpolation. Compare it with the electronic-only ranging residual before treating the plotted budget as an achievable floor.

The injected GW phase propagates with the same static science weights $(1 - AB)P_{ij}$, providing a predicted amplitude and phase for a specified waveform.

Draft note: Choose a named source or a specified monochromatic injection and give its predicted output amplitude and recovery criterion using the same delays as the noise calculation.

V. FUTURE EXTENSIONS

A. Injection of Secondary Noise

Test-mass displacement can be emulated by adding the corresponding phase terms M_{ij} from Section II to the synthesized links. Each simulated test mass must use the same displacement realization wherever it occurs, including the delayed occurrence in the reverse link. Independent noise added to every link would not reproduce these correlations. For a specified acceleration spectrum, the displacement ASD is obtained by dividing by $(2\pi f)^2$ over the analysis band. This injection would test the secondary-noise transfer functions as well as primary-noise suppression.

Independent phase perturbations can also be added after link synthesis to test a prescribed LISA-like readout-noise model. Lower optical power would test weak-light tracking and readout, but would not by itself remove the

correlations introduced by splitting one detector output across boards.

B. Time-Dependent Delays

The six-link design supports TDI combinations beyond the four-link Michelson considered here. Orbit-derived delays and Doppler shifts would extend the present static analysis to arm flexing and second-generation TDI. The analysis here uses phase data. For a time-dependent delay, differentiation gives

$$\frac{d}{dt}[\mathbf{D}_{ij}\delta\phi_j] = (1 - \dot{d}_{ij})\mathbf{D}_{ij}\delta\dot{\phi}_j. \quad (29)$$

Thus frequency-domain measurements require Doppler-weighted delay operators [16]. In this hardware, buffering tracked frequency and adding a nominal Doppler offset does not by itself supply the factor multiplying the fluctuating frequency.

Draft note: Check whether the firmware scales the delayed frequency fluctuations by $1 - \dot{d}_{ij}$. Implement this factor or quantify the deviation before claiming an orbit-consistent phase response. Use the represented optical frequency for the nominal Doppler shift, and rederive the board-jitter coefficients with the actual output frequency.

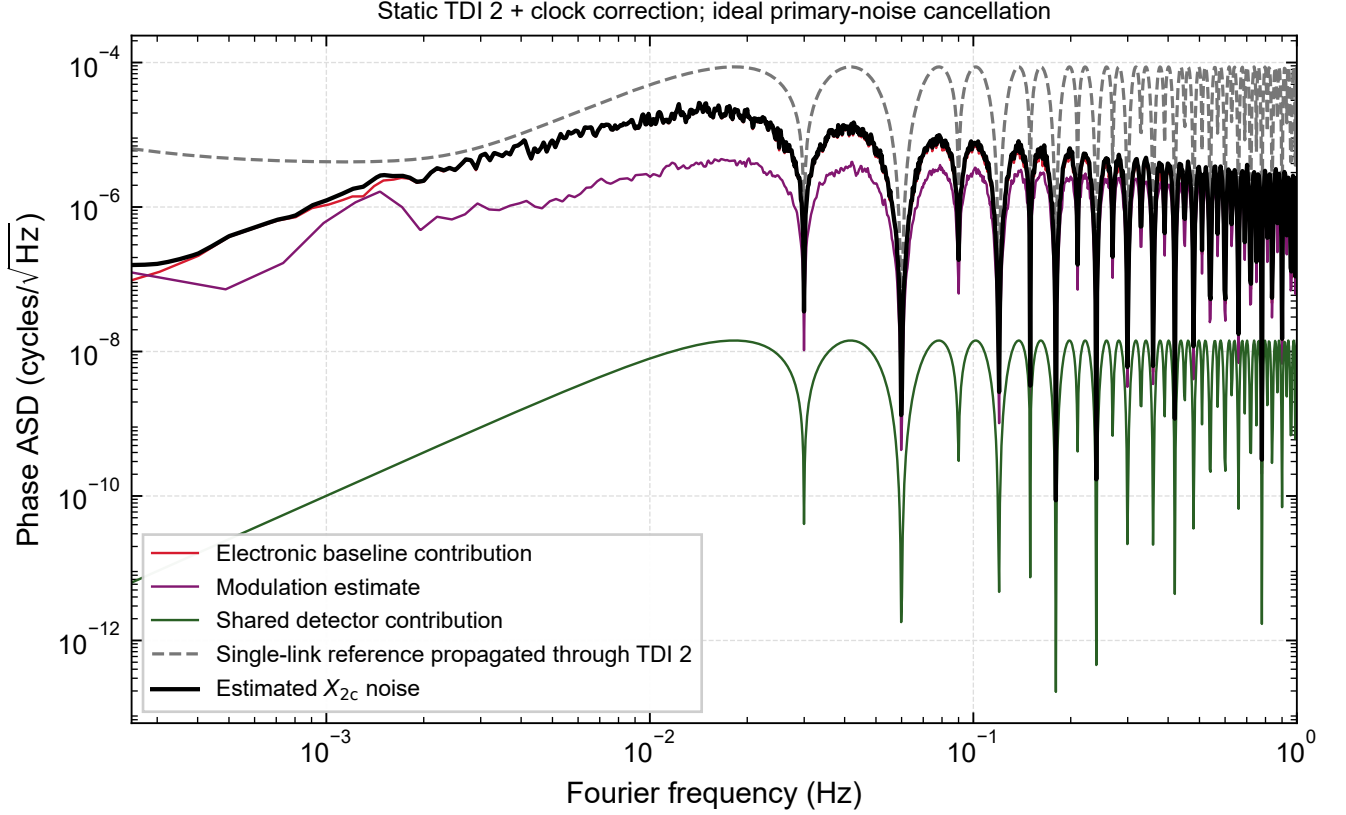
Draft note: For a full TDI-2 analysis, use ordered, nested delays and time-dependent carrier and sideband clock coefficients, then verify the corresponding clock correction. Give a TDI-1 flexing residual for the selected orbit and compare it with TDI-2; the static filter $(1 - AB)$ alone does not demonstrate flexing-arm cancellation.

Draft note: Specify delay-command update rates, accumulator precision, interpolation error and logging of realized delays. Check the maximum commanded delay against the 8.5 s memory limit. Describe time-frame alignment and whether processing delays are commanded, logged or fitted by TDI ranging, including clock and cable offsets.

Draft note: Develop and validate a frequency plan over the chosen orbit: all three output tones must remain trackable without zero crossings or overlap. A digital output offset changes sampling-clock and DAC-jitter couplings and must be included explicitly in both carrier and sideband coefficients; it is not automatically equivalent to an optical frequency-plan change.

C. Changes to the Clock-Transfer and Locking Configuration

Changing the modulation frequency requires separating the measured estimator into modulation, timing and readout contributions, as discussed in Section II C. A GHz implementation would also require a compatible optical and RF frequency plan. Modulating the reference



Modulation (SC1,2,3): 35, 36, 34 MHz; electrical carriers: 15, 10.9, 21.6 MHz
 Linear model: 4 mW/beam, $\mu_{\text{EOM}} = 0.53$; ADC covered by electronic baseline

FIG. 4. Estimated phase-noise ASD of the clock-corrected static Michelson observable X_{2c} . The black curve sums the electronic, assumed source-modulation and shared-detector contributions in quadrature. The calculation uses four links with 8.339 s delays, modulation frequencies (35, 36, 34) MHz, and the optical operating point of Section IV B. ADC noise is represented by the measured electronic baseline. The dashed comparison curve treats the single-link OMS-plus-acceleration reference as independent equivalent link noises propagated through the same TDI observable. Laser and common clock noise cancel ideally in this static estimate.

laser would allow corresponding optical sidebands to beat within the electronic input band.

A second optical source and modulator on each spacecraft, together with reference interferometers and back-links, would reproduce the local $\xi/z/\eta$ reduction and the β clock couplings omitted here [12]. Distinct synthesis chains and a local sideband comparison would also allow tests of differential tone-generation noise.

VI. CONCLUSION

We have presented the miniLISA design, which extends an electronic delay-line testbed with optical sources and independently referenced spacecraft measurements. Building on hardware demonstrations with electronic propagation delays [6, 9], it is intended to test laser-noise suppression, a reduced clock-transfer model, and GW signal recovery within the same measurement chain.

The estimate combines the measured electronic and modulation spectra with a prospective detector readout model, using the appropriate source and link transfer functions. Bright laboratory beams keep the predicted photodetector readout noise below the LISA science-interferometer allocations. Board timing introduces an additional coupling, including fluctuations of the effective delay, whose residual depends on differential board jitter and the delays used in processing. This component estimate provides a reference for the performance of the assembled optical testbed.

The next steps are to repeat the modulation-noise measurement, assemble the optical readout, and test noise suppression and GW recovery with the six-link setup. Varying delays and higher modulation frequencies would extend these tests.

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- [1] Benjamin P Abbott, Richard Abbott, Thomas D Abbott, Matthew R Abernathy, Fausto Acernese, Kendall Ackley, Carl Adams, Thomas Adams, Paolo Addresso, Rana X Adhikari, et al. Observation of gravitational waves from a binary black hole merger. *Physical review letters*, 116(6):061102, 2016.
 - [2] Monica Colpi, Karsten Danzmann, Martin Hewitson, Kelly Holley-Bockelmann, Philippe Jetzer, Gijs Nelemans, Antoine Petiteau, David Shoemaker, Carlos Sopuerta, Robin Stebbins, Nial Tanvir, Henry Ward, William Joseph Weber, Ira Thorpe, Anna Dauriskikh, Atul Deep, Ignacio Fernández Núñez, César García Marirrodiga, Martin Gehler, Jean-Philippe Hailain, Oliver Jennrich, Uwe Lammers, Jonan Larrañaga, Maike Lieser, Nora Lützgendorf, Waldemar Martens, Linda Mondin, Ana Piris Niño, Pau Amaro-Seoane, Manuel Arca Sedda, Pierre Auclair, Stanislav Babak, Quentin Baghi, Vishal Baibhav, Tessa Baker, Jean-Baptiste Bayle, Christopher Berry, Emanuele Berti, Guillaume Boileau, Matteo Bonetti, Richard Brito, Riccardo Busicchio, Gianluca Calcagni, Pedro R. Capelo, Chiara Caprini, Andrea Caputo, Eleonora Castelli, Hsin-Yu Chen, Xian Chen, Alvin Chua, Gareth Davies, Andrea Derdzinski, Valerie Fiona Domcke, Daniela Doneva, Irna Dvorkin, Jose María Ezquiaga, Jonathan Gair, Zoltan Haiman, Ian Harry, Olaf Hartwig, Aurelien Hees, Anna Heffernan, Sascha Husa, David Izquierdo, Nikolaos Karnesis, Antoine Klein, Valeriya Korol, Natalia Korsakova, Thomas Kupfer, Danny Laghi, Astrid Lamberts, Shane Larson, Maude Le Jeune, Marek Lewicki, Tyson Littenberg, Eric Madge, Alberto Mangiagli, Sylvain Marsat, Ivan Martin Vilchez, Andrea Maselli, Josh Mathews, Maarten van de Meent, Martina Muratore, Germano Nardini, Paolo Pani, Marco Peloso, Mauro Pieroni, Adam Pound, Hippolyte Quelquejay-Leclere, Angelo Ricciardone, Elena Maria Rossi, Andrea Sartirana, Etienne Savalle, Laura Sberna, Alberto Sesana, Deirdre Shoemaker, Jacob Slutsky, Thomas Sotiriou, Lorenzo Speri, Martin Staab, Danièle Steer, Nicola Tamanini, Gianmassimo Tasinato, Jesus Torrado, Alejandro Torres-Orjuela, Alexandre Toubiana, Michele Vallisneri, Alberto Vecchio, Marta Volonteri, Kent Yagi, and Lorenz Zwick. LISA definition study report, 2024. Version Number: 1.
 - [3] Jean-Baptiste Bayle and Olaf Hartwig. Unified model for the LISA measurements and instrument simulations. *Physical Review D*, 107(8):083019, 2023.
 - [4] Gregory Weaver, Jeffrey Garstecki, and Samuel Reynolds. The performance of ultra-stable oscillators for the gravity recovery and interior laboratory (grail). In *Proceedings of the 42nd Annual Precise Time and Time Interval Systems and Applications Meeting*, pages 369–380, 2010.
 - [5] Massimo Tinto, F. B. Estabrook, and J. W. Armstrong. Time-delay interferometry for LISA. *Physical Review D*, 65(8):082003, 2002. Publisher: American Physical Society.
 - [6] Shawn J. Mitryk, Guido Mueller, and Josep Sanjuan. Hardware-based demonstration of time-delay interferometry and TDI-ranging with spacecraft motion effects. *Physical Review D*, 86(12):122006, 2012.
 - [7] Glenn de Vine, Brent Ware, Kirk McKenzie, Robert E. Spero, William M. Klipstein, and Daniel A. Shaddock. Experimental demonstration of time-delay interferometry for the laser interferometer space antenna. *Physical Review Letters*, 104:211103, 2010.
 - [8] M. Laporte, H. Halloin, E. Bréelle, C. Buy, P. Grüning, and P. Prat. Status of the LISA On Table experiment: a electro-optical simulator for LISA. *Journal of Physics: Conference Series*, 840:012014, 2017.
 - [9] Reid Ferguson, Olaf Hartwig, and Guido Mueller. Applying tdi to hardware-simulated data. *arXiv preprint arXiv:2505.22579*, 2025.
 - [10] Gerhard Heinzel, Juan José Esteban, Simon Barke, Markus Otto, Yan Wang, Antonio F Garcia, and Karsten Danzmann. Auxiliary functions of the **LISA** laser link: ranging, clock noise transfer and data communication. *Classical and Quantum Gravity*, 28(9):094008, 2011.
 - [11] Gerhard Heinzel, Javier Álvarez Vizoso, Miguel Dovale-Álvarez, and Karsten Wiesner. Frequency planning for LISA. *Physical Review D*, 110(4):042002, 2024.
 - [12] Markus Otto, Gerhard Heinzel, and Karsten Danzmann. Tdi and clock noise removal for the split interferometry configuration of lisa. *Classical and Quantum Gravity*, 29(20):205003, 2012.
 - [13] Massimo Tinto and Olaf Hartwig. Time-delay interferometry and clock-noise calibration. *Physical Review D*, 98(4):042003, 2018.
 - [14] Olaf Hartwig and Jean-Baptiste Bayle. Clock-jitter reduction in LISA time-delay interferometry combinations. *Physical Review D*, 103(12):123027, 2021. Publisher: American Physical Society.
 - [15] Stanford Research Systems. FS725 rubidium frequency standard: specifications. <https://www.thinksrs.com/products/fs725.html>. Accessed 22 September 2026.
 - [16] Jean-Baptiste Bayle, Olaf Hartwig, and Martin Staab. Adapting time-delay interferometry for LISA data in frequency. *Physical Review D*, 104(2):023006, 2021. Publisher: American Physical Society.